TUX_3

Our YouTube

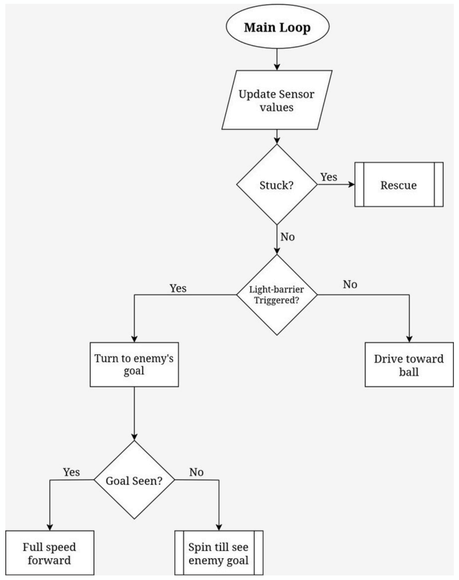
Training video

Our GitHub

Team

From left to Right:
Alexander Belov (Software and Team Captain)
Toni Schürmann (Software)
Max Harting (Hardware)
Henning Baumgartner (Hardware)

Software



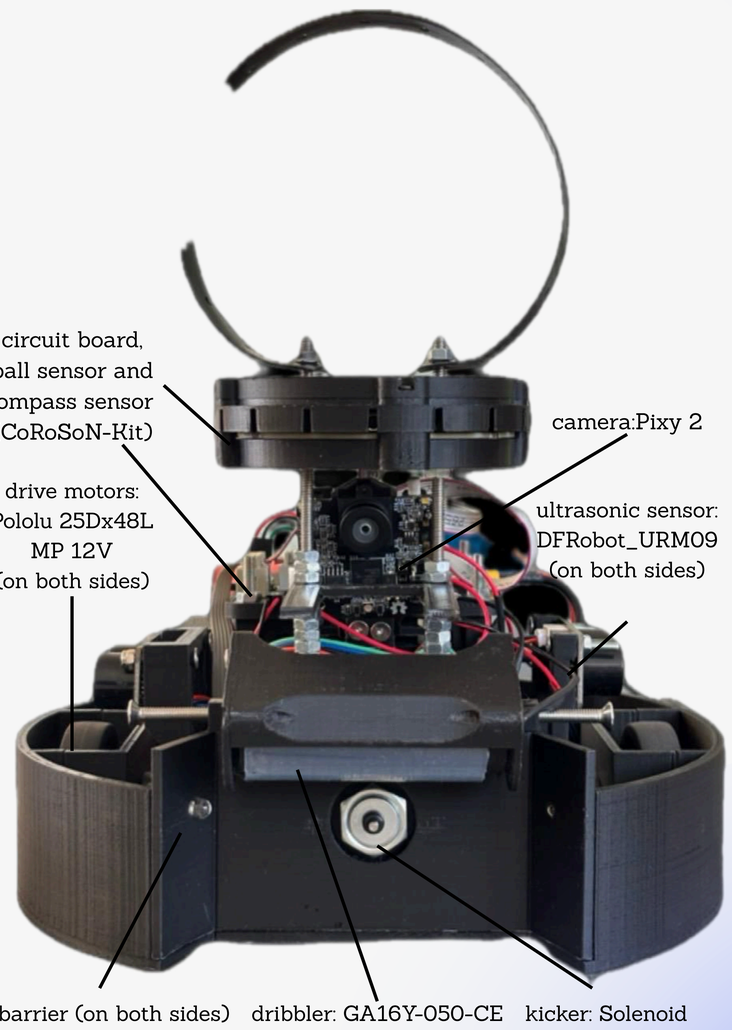
Abstract

We are four students from the Gymnasium Rahden, a small town in North Rhine-Westphalia near Bielefeld. We go to two different classes. Despite this separation at school, we are united by a close friendship and a shared interest in robotics. Almost two years ago, we joined our school's robotics club. Since then, robotics has become an integral part of our lives, as we not only work in the club but also dedicate time on weekends and during holidays.

After an initial training phase in which we learned basic programming commands, we developed a strong interest in the practical implementation of technical concepts. During this time, we also learned a great deal about how to divide tasks effectively. As we progressed, we designed and programmed our first LEGO robot. In 2025, we initially competed as two separate teams at the Berlin Open. There, we achieved remarkable results: one team secured second place, while the other finished fourth. These performances qualified us for the German Open 2025, where we placed eighth and thirteenth.

In this school year, we expanded our skills particularly in the field of additive manufacturing (3D printing) as well as in hardware development. In addition, we decided to merge our two teams in order to achieve a better overall result and to combine our strengths. This collaboration led to the construction of a high-performance robot for the Berlin Open 2026, where we once again achieved second place.

For the German Open 2026, we worked even harder compared to the previous year and are now aiming to qualify for the Europe Open in Vienna. Regardless of the outcome, we are proud of what we have already achieved.



Hardware

The robot was collaboratively developed and built by all team members. The goal was to create a functional and competitive 1-vs-1 entry robot capable of autonomously collecting the ball, controlling it, and shooting it into the goal.

The system includes a curcit board, a ball sensor for object detection, two drive motors, three ultrasonic sensors for distance measurement, a dribbler for ball control, a PixyCam for goal detection, two lightbarriers for object detection and a kicker for precise shooting. A bill of materials for the main components can be found at the roboter description .

The construction process was carried out chronologically in several stages. First, the Chase and mounting brackets were manufactured using 3D printing. The material we are using for our robot is PLA+. Afterwards, we installed the ball sensor, two ultrasonic sensors, the motors, two lightbarriers and the dribbler. Following our participation in the Berlin Open 2026, the system was further improved: a third ultrasonic sensor, a PixyCam, a kicker, and a new Chase were added to enhance overall performance.

These were the key decisions we made during the development of our robot:
- A front-facing _/-shaped opening improves ball intake and enables faster and more stable ball control through the dribbler.
- The retrofitted kicker addresses previously identified weaknesses in goal finishing and now allows for direct and controlled shooting.
- Two laterally positioned ultrasonic sensors ensure more precise alignment with the opponent's goal and support stable driving direction.

The total development and construction time was approximately ten months, the overall cost of the robot amounts to €310, and we used the programming language C++ for programming.